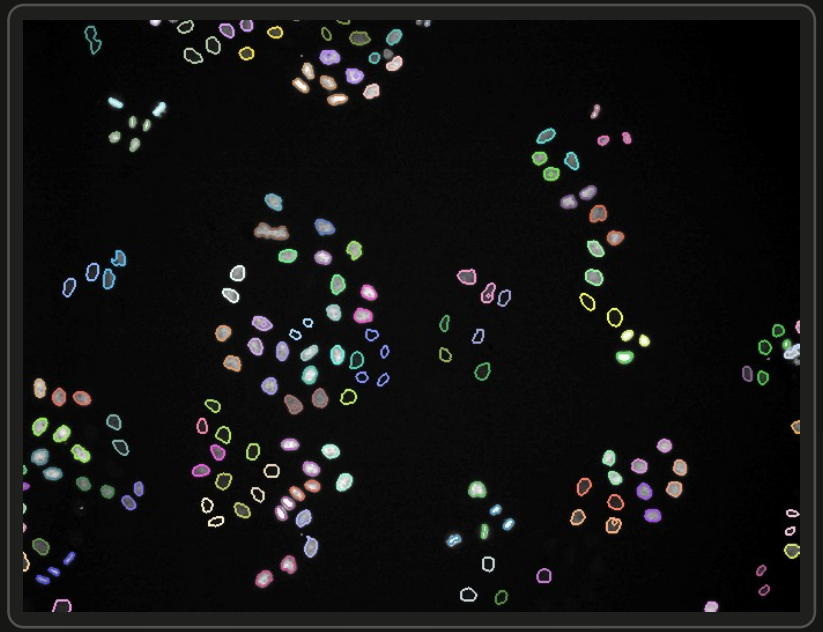
GLOBIAS 2026 · GROUP 5 · SESSION 06
BENCHMARKING

Can an AI agent run, and rerun, a bioimage analysis?

One 2014 ilastik workflow, five Claude sessions



Frame 40 of the MitoCheck movie, each lineage outlined in its own colour

What happened here

AN AGENT SESSION

Claude gets a computer. It runs commands, installs software, reads and writes files.

It works in a loop. Command → error → try again, on its own, until done or stuck.

A person watches. They answer questions and can step in at any time.

THE WORKFLOW: ILASTIK'S MITOCHECK EXAMPLE

Raw movie
53 frames of dividing nuclei

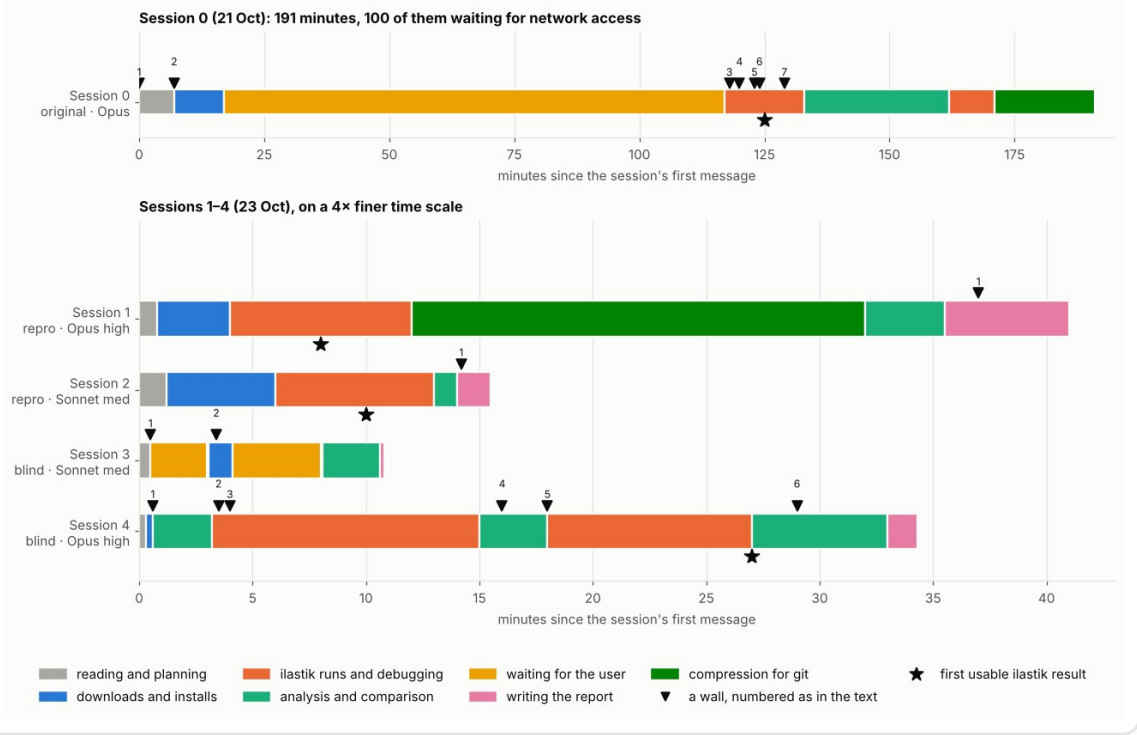
Pixel classification
a probability map per frame

Tracking
lineages: nucleus + descendants



Frame 26, ilastik 1.3.3post3 output from Run 0. Right: each lineage in its own colour; grey objects were detected but not tracked.

Run 0: seven walls to the first result



191 min in total

100 min waiting for network access

7 walls

- 1.4.1 won't install → 1.5.0a3 pre-release → 1.3.3post3
- Two walls needed a small fix script
- Two unchecked claims, caught on a second look

Runs 1 and 2: with the notes, no help needed



Orange: hit. Blue: already fixed by Run 0's scripts. Green: avoided another way. Grey: not met.

0 help needed

≈15 / 41 min, Sonnet / Opus

0 of 7 walls met

1.5.0a3: tracking identical to Run 0's, pixel for pixel

1.3.3post3: not exactly repeatable. Opus rerun: two lineages swapped numbers, 113 pixels changed object; Sonnet rerun: 50 pixels differed

Runs 3 and 4: without the notes, a different experiment

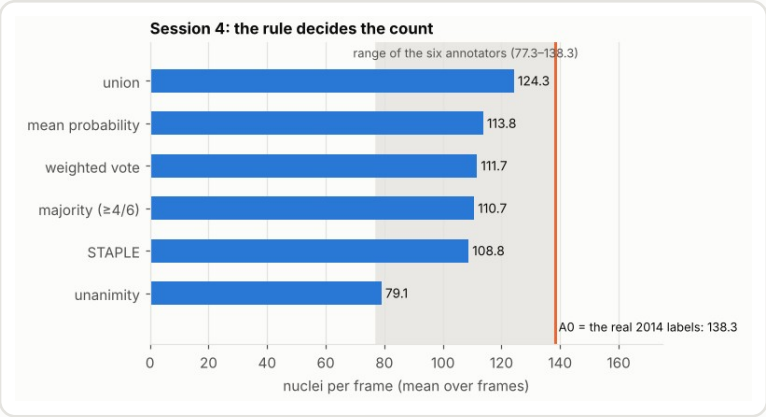
	Run 3 · Sonnet 11 min · asked twice	Run 4 · Opus 34 min · no questions
LOOKED FIRST	The computer → “ilastik needs a window”	The download page → ilastik 1.4.2, run headless
ANNOTATORS	One person's labels, resampled into five classifiers	Real 2014 labels as A0, plus five invented styles
RULES COMPARED	5 rules on 7 frames	7 rules over the whole movie

CONCLUSION

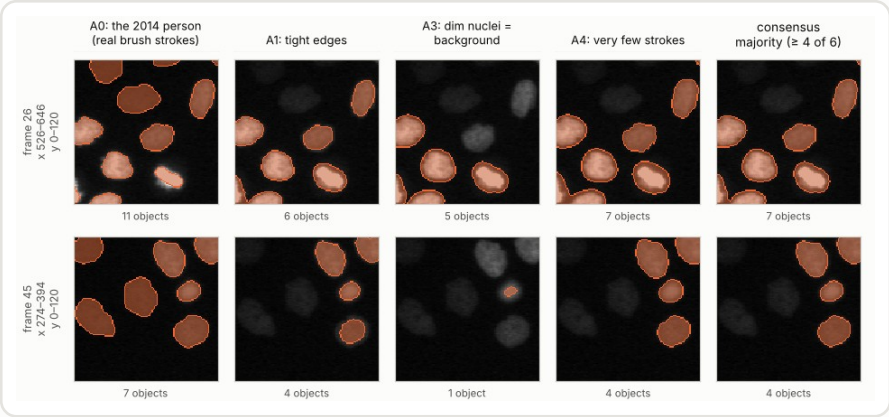
Both: consensus is not sufficient to define ground truth

The rule decides the count

In the biased pool, majority vote landed at **84 nuclei per frame** against A0's **138**, and the agreement scores ranked A0 the least reliable.



Run 4, main pool: nuclei per frame under each rule (blue), the six annotators' range (grey), A0 (orange).



Run 4, frames 26 and 45: A1, A3 and A4 miss most of the dim nuclei that A0 finds; majority vote drops them.

What the five runs show

Documentation turns a problem into a recipe.
With Run 0's notes, the two reproductions met none of its seven walls. Neither needed help.

The first judgement decides the rest.
Run 3 looked at the computer, concluded ilastik could not run, and built everything on that.

Agents choose the experiment when the task is open.
Both blind runs answered with a consensus experiment of their own design.

Every session overstated something.
A second look, by a person or by another session, caught each one.

Numbers can mislead.
99.99% of pixels match the 2014 map, but an empty image scores 95%; nuclei overlap 99.8%.

globias-2026-group5.github.io

Analyses run by Claude (Anthropic) agent sessions; data courtesy of the MitoCheck project; example and projects from ilastik.